

Banks' Incentives and Inconsistent Risk Models

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This paper investigates banks' incentive to bias the risk estimates they report to regulators. Within loan syndicates, we find that banks with less capital report lower risk estimates. Consistent with an effort to mitigate capital requirements, the sensitivity to capital is robust to bank fixed effects and greater for large, risky, and opaque credits. Also, low-capital banks' risk estimates have less explanatory power than those of high-capital banks with regard to loan prices, indicating that their estimates incorporate less information. Our results suggest banks underreport risk in response to capital constraints and highlight the perils of regulation premised on self-reporting. (*JEL* G21, G28)

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The regulation and supervision of the financial industry is increasingly reliant on information produced by regulated entities. For example, banks' internally generated risk estimates are now being used to determine capital requirements. However, in the vein of Goodhart's law or the Lucas critique (Lucas 1976), the quality of this information is not invariant to its use. In the case of banks, capital standards make reported risk costly for institutions, a cost that encourages them to seek relief by reporting lower risk estimates. In this paper, we examine this eventuality by investigating differences in banks' risk estimates across commonly held credits.

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We do so using the risk metrics banks report to regulators for syndicated loans. In a syndicated loan, multiple institutions lend to a single borrower under a common loan contract. We observe that banks report substantively different risk estimates for the same credit and that these differences are systematic: some banks consistently deviate from their peers. More importantly, we find that these differences are correlated with banks' capital ratios: banks with less capital report that their loan investments are less risky. Lastly, we find that the interest rates that low-capital banks charge on loans are poorly explained by their risk estimates. The confluence of these results is consistent with an incentive to mitigate capital constraints, but not easily explained by alternative hypotheses that we consider.

When the Basel I Accord was introduced in 1998, it was praised for incorporating risk into assessments of bank capital. Under Basel I, on-balance sheet assets are bucketed into broad risk categories and each category is assigned a fixed risk weight that determines the amount of capital banks need to set aside. This simple approach discourages holding relatively safe assets in a category and it does not incorporate "soft" information about creditors that is costly to produce but central to relationship lending (Stein 2002).

Pursuant to advances in risk modeling, the Basel Committee proposed a new capital adequacy framework in 2004, Basel II, that sought to address these criticisms. A key innovation in Basel II was that it allowed banks to use an internal ratings-based approach to determine their required capital levels. Under this approach, the risk weight of a loan is a function of the bank's internally generated estimates for the borrower's probability of default (PD), the bank's loss given default (LGD), and the bank's exposure at default (EAD).

The internal ratings-based approach was lauded because it built on banks' own information and made capital standards more risk sensitive. But this conclusion overlooks the wisdom of Goodhart and Lucas, because it assumes banks will produce and share accurate risk estimates without regard for the policy-induced outcome. Disclosing high risk estimates is costly to banks as it increases the amount of capital they must set aside for regulatory purposes.¹ Even when risk estimates are supervised, the soft information produced by banks is by its very nature difficult to verify, creating significant scope for discretion. Moreover, there is no specified penalty for poor ex post model performance. As a result, banks have both the incentive and the opportunity to selectively incorporate private information.² This is problematic because it

¹ This cost is one reason the "unraveling result"—that firms disclose private information to maximize their value—does not hold in this context (Grossman 1980; Grossman and Hart 1980; Milgrom 1981; Milgrom and Roberts 1986). For models that rationalize partial disclosure of private information, see Verrecchia (1983), Dye (2000), Jung and Kwon (1988), Hughes and Pae (2004), and Kahn and Santos (2006).

² While we typically refer to firms' incentives, the incentive to reduce required capital is in fact transmitted throughout these institutions to managers via internally allocated limits on capital and compensation contracts that reward higher returns.

can misrepresent banks' riskiness, undermining capital regulations and creating competitive inequities.

It is empirically difficult to detect systematic differences in banks' risk estimates. Internal risk estimates are typically not reported, and, when they are, they are at the portfolio level, which presents identification challenges. To identify a potential capital incentive, we focus on internal ratings for commonly held credits that we can compare across banks.

We exploit the unique features of the Shared National Credit (SNC) program to investigate whether there are systematic differences in banks' reported risk metrics under the internal ratings-based approach. Beginning in 2009, banks adopting Basel II must provide risk metrics necessary for calculating the risk weight of the loan, including the PD, LGD, and EAD for each credit exposure. Therefore, we observe different banks' estimates for the *same* credit at the *same* point in time. We can calculate differences in these ex ante risk estimates to determine whether there are inconsistencies across banks.

Our sample includes fifteen banks over the period 2010Q2 to 2013Q3. We restrict our analysis to credits with at least two reporting banks. In that period, we observe over 7,500 credit facilities, reflecting over 3,500 distinct borrowers. While we consider the range of risk metrics, the focus of our analysis is on PDs because the probability of default is borrower based and independent of bank-specific policies including hedging strategies.

We find that internally generated risk estimates, and their implied risk weights significantly vary across syndicate-member banks. We discover some banks report risk metrics systematically above the average of the syndicate and others systematically below. Our results imply that some banks are able to hold less capital for a common set of credits because they report lower risk estimates.

Inconsistent risk estimates are not surprising. In the spirit of the Basel reforms, some banks might collect and incorporate private information into their risk estimates and as a consequence produce risk estimates that are unique relative to their peers. But private information does not easily explain *systematic* differences that consistently deviate in one direction relative to other banks. Even if some banks are more adept at producing private information, this information should not reduce the riskiness of loans vis-à-vis other banks; instead, it should improve the accuracy of risk estimates. Significant bank effects demonstrate that bank-level factors contribute to inconsistent risk estimates.

We investigate these differences in risk estimates further and find that on average banks with less regulatory capital report lower PDs and lower risk weights than do their peers. This is true at the credit level and when we aggregate loans into portfolios.³ At the portfolio level, the difference in PD is as large as

³ Prior research has documented that banks evergreen to minimize capital charges from nonperforming loans (e.g., Peek and Rosengren 2005). Our evidence suggests another mechanism by which firms can reduce capital charges from all loans, including distressed ones.

80 bps between low- and high-capital banks. For a typical loan, this could vary risk-weighted assets by 20%, with a corresponding impact on required capital.

The positive correlation with capital is consistent with an incentive to improve regulatory capital. However, such a correlation might also result from heterogeneity in risk perception or risk attitude. If a bank's overall perception of risk is lower, this would also inform their leverage (i.e., their Tier 1 ratio), resulting in a correlation between low capital and low risk estimates. To account for persistent bank characteristics, we repeat the analysis in the presence of bank fixed effects. We find that the positive relation between PD bias and bank capital is robust to this alternative specification. We also use the overall riskiness of the loan portfolio as a control for risk perception. We find that riskier portfolios are not associated with lower risk estimates and do not explain the positive association between differences in PDs and capital.

An alternative concern is that some banks may use private information to choose the loans they hold. As a result, the more selective banks will invest in loans for which they have lower risk estimates relative to other banks. To account for this explanation we include the participation rate of banks in the universe of credits. If banks only select the relatively less risky credits from an unconditional distribution that resembles their peers, then their participation rate should be lower. Indeed, we find that banks with a lower participation rate tend to report less risk, but that participation does not explain the relationship with measures of capital.

Throughout our analysis we control for the role of the agent bank, that is, the primary lead arranger of the syndicated loan; however, the literature on syndicated loans has highlighted the key role of lead arrangers. The broader set of lead arrangers may behave differently because of their access to information or the frequency with which they update their risk estimates. We conduct several robustness tests that consider the relation between risk metrics conditional on the role of the bank in the syndicate, the freshness of the risk metrics, and the age of the credit. We find that capital levels and PDs are positively correlated for lead arrangers and also for non-lead-participating banks alleviating concerns that our results are driven by the role of banks in the syndicate. Our primary findings are present among the set of freshly updated risk estimates and in the first few quarters of a credit's life, ruling out the hypothesis that our results are driven by heterogeneous updating of risk estimates by banks.

We also examine whether the sensitivity of PDs to capital is greater for certain types of credits. We find that the downward bias for low-capital banks is greater for riskier credits and for credits that are drawn which have greater exposure at default. Both findings are consistent with a regulatory incentive: Risky credits have more scope for disagreement or discretion and drawn credits have a larger impact on the RWA of a loan portfolio. Additionally, we find that the relation between bias and capital is greater for private borrowers where banks have more discretion over the information used to estimate the probability of default. The attenuation we find for public borrowers is consistent with the hypothesis

that private information is the channel by which banks are able to influence risk metrics. These facts are not easily explained by loan selection or risk perception explanations, but are consistent with an effort to reduce regulatory capital constraints.

To further disentangle competing stories, we investigate the pricing of loans using internal risk estimates. An effort to minimize regulatory capital that relies on the discretionary production of risk estimates will result in downward biased risk estimates that are noisier with respect to prices. The results show that the PDs reported by low-capital banks have less explanatory power than those reported by high-capital banks with respect to loan spreads. Thus, low-capital banks report both lower risk estimates as well as risk estimates that incorporate less information than their peers. These two findings are consistent with a capital incentive that distorts some risk estimates but are not explained by lower risk perception, for in the latter there is no reason to expect a cross-sectional difference in the explanatory power of reported risk metrics on loan interest rates.

While there is no perfect test of an institution's motives, on whole our results are consistent with an effort by low-capital banks to mitigate capital constraints and are difficult to reconcile with alternative theories. Regardless of the underlying cause, our findings illustrate a crucial weakness in the use of self-reported risk estimates: the least well-capitalized banks report risk estimates that are lower than their peers. Hence, those banks whose buffers to adverse economic shocks are already low are less well-capitalized than they appear.

Our results highlight the need for a robust supervisory program in a regulatory regime that relies on self-reporting. In the United States, the quality of internally generated risk estimates is one criterion that determines whether banks can fully adopt Basel III for capital purposes. However, it is also crucial to introduce mechanisms that entice banks to report unbiased risk metrics even after receiving such approval. To this end, the insights of Kaplow and Shavell (1994) could be useful. They show that in models of self-reported behavior the expected punishment must exceed the gains from inaccurate reporting. In designing these mechanisms, it would be important to consider how formal comparisons between banks may influence their risk estimates. Analogous to the incentives described in the Keynesian "beauty contest"⁴ or Scharfstein and Stein (1990), banks may choose to disclose information not based on what they think the risk of the loan is, but rather on what they believe the other banks that also own the loan believe the risk is.

There are additional challenges: benchmarking risk estimates is difficult when assets are not owned across banks and there are constraints on the type of

⁴ See Keynes (1936, chap. 12) for a presentation of the general idea. Entrants are asked to choose the six prettiest faces from a hundred photographs, with the contestant choosing the most popular face receiving a prize. Keynes writes: "We have reached the third degree where we devote our intelligences to anticipating what average opinion expects the average opinion to be."

information supervisors can verify. Ultimately, it is not feasible to monitor the quality of the inputs to each and every risk estimate. Therefore, our findings lend support to proposals to complement the existing risk-based capital standards with a simple leverage ratio. Doing so would limit the impact of underreported risks on bank capital (Basel Committee on Banking Supervision 2014; Acharya, Engle, and Pierret 2014).

Our paper reinforces the emerging literature on the inconsistencies of internal risk models across banks.⁵ RMA Capital Working Group (2000) and Firestone and Rezende (2016) document similar heterogeneity by banks participating in syndicated loans. Unlike these studies, which rely solely on cross-sectional differences, our data forms a quarterly panel over a 3-year period. More importantly, our focus is not limited to identifying inconsistencies across banks; we are also interested in understanding the source of these inconsistencies.

Two additional related papers are Begley, Purnanandam, and Zheng (2017) and Behn, Haselman, and Vig (2014). Begley, Purnanandam, and Zheng (2017) document that the frequency of value-at-risk violations in bank trading books is correlated with bank capital. Using German data, Behn, Haselman, and Vig (2014) find that banks report lower PDs for loans whose capital charges were determined by internal models when compared to safer loans originated under the standardized approach. The former study relies on comparisons across disparate portfolios of assets and the latter compares common borrowers but across different credits. In contrast, we are able to make cross-bank comparisons using a portfolio of commonly held credits at the same point in time, significantly reducing the potential alternative explanations for our findings. Further, both of these papers consider *ex post* measures of bias, whereas we focus on *ex ante* measures because these are the measures that enter into regulatory capital calculations.

Finally, our paper contributes to the literature on the role incentives play in the production of risk estimates (e.g., Rajan, Seru, and Vig 2010). Prior work has suggested incentives distorted estimated risks in the mortgage securitization market (Rajan, Seru, and Vig 2015), noting that the nature of the information (soft vs. hard) is important. Our paper documents evidence of this behavior in the context of banking regulation. For example, we discover estimates for loans with more soft information (i.e., nonpublic firms) are more sensitive to measures of capital constraints.

1. Data and Sample Summary

In this section we describe the internally generated risk metrics introduced by Basel II and provide background information on our primary data source, the

⁵ Carey (2002) and Jacobson, Linde, and Roszbach (2006) show that internal risk metrics are not consistent across banks by comparing the ratings different banks assign to loans of a given borrower. Financial Services Authority (2012) observes bank-level differences between risk metrics, PD and LGD, of a hypothetical common portfolio of credits.

Expanded Shared National Credit Program. The final portion of this section describes our sample.

1.1 Basel II: Advanced internal ratings-based approach

Since Basel I, risk-weighted assets (RWA) have been a key component of bank regulatory ratios. The most prominent example is the Tier 1 capital ratio which is the ratio of a bank's core equity capital to RWA. Under Basel I, assets are prescribed a risk-weighting based on five distinct risk buckets: 0%, 10%, 20%, 50%, and 100%. The risk-weighted sum of exposures equals the bank's total RWA that are used to calculate various regulatory ratios.

A drawback of this approach is that assets with different risks are assigned the same risk weight. Basel II regulations seek to mitigate this problem by introducing an alternative capital framework. Basel II, and its successor Basel III, allow banks to use either the standard approach or the Advanced Internal Ratings-Based Approach (AIRB). The standard approach made capital requirements on corporate loans dependent on the rating of the borrower where unrated borrowers were assigned a fixed risk weight of 100%. In contrast, the AIRB approach allows banks to estimate the risk weight of a loan using their internal models.

In the AIRB approach, there are four self-reported components of the corporate loan risk-weight calculation. The first is the borrower's probability of default (PD). Banks estimate the PD using historical models of default that relate the characteristics of borrowers to default frequencies.

The second self-reported component of risk weights is loss given default (LGD). LGD is reported before and after credit-risk mitigants (CRM), which can consist of collateral, guarantees, and credit derivatives. In contrast to PDs, which are borrower specific, LGDs tend to be based on bank-specific experiences with particular industries and types of collateral. The third component is the time to maturity of the loan. PD, LGD after CRM, and maturity are the inputs to a prescribed equation that calculates the risk weight of the loan (see Basel Committee on Banking Supervision 2006). The fourth and final estimate is exposure at default (EAD), which reflects the dollar amount outstanding at default. The risk weight is multiplied by the EAD to calculate the risk-weighted asset value in dollars.

1.2 The expanded Shared National Credit Program

Our objective is to compare reported risk estimates across banks. We exploit syndicated loan data from the Shared National Credit (SNC) Program. In a syndicated loan, at least two institutions agree to provide credit under a common loan agreement. One of the lenders, known as the agent or lead, acts as the intermediary between the loan syndicate and the borrower. The agent will negotiate the loan and administer payments to and from the borrower. In exchange for their services, agents receive a small fee (see, e.g., Dennis and Mullineaux 2000 for additional details on syndicated lending).

The SNC Program is administered by The Federal Reserve System (FRS), the Office of the Comptroller of the Currency (OCC), and the Federal Deposit Insurance Corporation (FDIC). The program collects data on an annual basis from syndicated loan agents on any term loan or revolver for which the aggregate value is \$20 million or more and which is shared by, or sold to, two or more federally supervised institutions. Agents report detailed data on the loan and syndicate, including the composition of the syndicate, the type of loan, and the borrower.⁶

Beginning in 2009, the SNC Program was expanded. Banks adopting the AIRB approach were designated as “expanded” reporters and were required to begin reporting their participation in credits quarterly.⁷ Expanded reporters must also provide their internal risk metrics for these credits. If the bank is using or preparing to use the AIRB to determine Basel II capital adequacy, it reports the risk metrics necessary to calculate their Basel II risk weights. While internal credit ratings are difficult to compare across banks because they lack a clear correspondence (both in levels and economic meaning), Basel II metrics share a common definition as outlined by regulations.

To be approved for this approach, banks must enter a “parallel-run” period, during which they remain subject to categorical risk-based capital rules until the regulator approves their transition to using the AIRB approach. Our sample includes banks that have already been approved to use AIRB for capital purposes as well as those undergoing a parallel run.⁸

Importantly, it is unlikely that banks anticipate these comparisons during our sample period. The focus of the SNC program has historically been on internal ratings rather than Basel II risk weights and at the time of our study there were no other papers that use SNC data to make within syndicate comparisons of risk estimates. Nevertheless, any expectation that bias is observed and subject to a penalty would mitigate banks’ incentives to bias estimates.

1.3 Sample summary

Using SNC program expanded reporters, we construct a panel of commonly held credit-bank-quarters where a credit is a syndicated loan. The identification of differences requires that multiple institutions report risk metrics hence we focus on those credits where at least two banks report PDs.⁹ In addition, credits

⁶ The SNC data were solely processed within the Federal Reserve for the analysis presented in this paper. For other studies that use data from the SNC program, see Bord and Santos (2012, 2014) and Mian and Santos (2017).

⁷ Basel II adoption is mandatory for large internationally active banking organizations (so-called “core” banking organizations with at least \$250 billion in total assets or at least \$10 billion in foreign exposure) and optional for others.

⁸ While parallel-run banks are not officially basing RWA on the internally generated risk metrics, they are still incentivized to use statistical models and methods that will ultimately result in a more favorable level of RWA. Our primary results are qualitatively similar across the two groups.

⁹ We exclude observations where banks substitute the PD of a guarantor for the PD of the actual borrower to ensure PDs refer to a common entity.

are limited to term loans and revolvers and exclude facilities designated as held for sale since banks are not required to report risk metrics for the latter. Lastly, we exclude credits that are at least 90 days past due, as AIRB may no longer reflect the required capital held against the position. The sample of commonly held credits consists of 7,606 distinct credit facilities, representing 3,636 unique borrowers over the fourteen quarters from 2010Q2 to 2013Q3.

Our analysis considers both the level of risk metrics, the log of risk metrics, and deviations from peer averages within a syndicate. Levels like PD or functions of PD, such as risk weights, exhibit extreme skewness driven by both the bounded nature of PDs and their clustering near zero (see Internet Appendix Figure IA1 for the distribution of PDs). To reduce the influence of these outliers, we trim the risk metrics that use PD as an input at the top 2% of their distribution. For PD this equates to trimming observations with an estimated default rate of greater than 20%. When extended to deviations, the skewness results in extreme outliers, both positive and negative. For these, we trim the sample at the top and bottom 1%. Risk metrics that are independent of PD and log based measures do not exhibit as dramatic a departure from normality and are not trimmed (see Internet Appendix Figure IA2 for the distribution of the natural log of PDs). We demonstrate throughout the paper that our findings are generally robust across trimmed risk metrics and log formulations.

Table 1 summarizes the properties of the sample across credits and banks. The average (median) number of syndicate participants is 14.7 (11). 16% of the credit-quarters are term loans with the complement being revolvers. On average we observe 3.4 expanded reporters per facility (where the minimum is two based on sample construction). 43% of the facilities are with public borrowers and 8% are “new” loans originated in the most recent quarter. The average credit in the sample is nine quarters old.

The middle portion of Table 1 outlines the relevant sample of bank-quarters. There are fifteen banks that report PDs, but for confidentiality purposes we cannot identify these institutions. The panel of reporting banks is unbalanced. In the initial quarter, 2010Q2, nine banks report and over time additional banks enter the sample with the final bank entering the sample in the first quarter of 2013. If the high-holder is a U.S. Bank Holding Company (BHC), bank financial data is sourced from the FR Y-9C regulatory reports. If the high-holder is a foreign bank, Tier 1 capital and assets are obtained from the FR Y-7Q regulatory filing and other financial data are derived from public filings with adjustments made to account for differences in accounting standards.¹⁰ The average number of commonly held credits for a bank in a particular quarter is 835 and the median is 491. The participation rate reveals that an average bank participates in 19% of outstanding credits. More than half of bank-quarters are banks with headquarters outside the United States.

¹⁰ For derivatives, IASB standards permit less balance sheet offsetting than FASB, resulting in larger balance sheets, all else equal.

Table 1
Summary statistics for credits and banks

	N	Mean	Median	SD
Credit-quarter:				
Commitment size (\$ mm's)	42,636	502.0	250.0	824.9
Utilized (\$ mm's)	42,636	117.9	37.7	238.3
Drawn (%)	42,636	38	23	39
Participants	42,636	14.7	11.0	26.2
Reporting participants	42,636	3.4	3.0	1.7
Term loan dummy	42,636	16%	0%	36%
Public borrower	42,636	43%	0%	50%
Age (Quarters)	42,636	8.6	6.0	9.1
New	42,636	8%	0%	28%
Average <i>PD</i> (%)	42,636	1.91	0.57	3.78
Average risk-weight (%)	42,600	69.0	63.0	37.0
Bank-quarter:				
Log(Assets)	174	21.0	21.3	0.8
Tier 1 ratio (%)	174	13.3	12.7	2.4
Tier 1 leverage (%)	174	5.0	4.7	2.2
ROE	174	7.3	7.2	5.7
RWA/Assets	174	39.3	31.6	19.9
Common credits	174	833.8	491.0	748.2
Participation rate (%)	174	19	11	18
Foreign (%)	174	58	100	49
Average <i>PD</i> (%)	174	1.59	1.55	0.58
Bank-credit-quarter				
Agent bank (%)	145,160	25	0	44
Share of credit (%)	145,160	13	10	11
<i>PD</i> (%)	142,978	1.21	0.38	2.33
<i>LGD</i> _{Bef} (%)	136,278	36.4	38.0	11.6
<i>PD</i> * <i>LGD</i> _{Bef} (%)	137,231	0.38	0.13	0.71
<i>LGD</i> _{After} (%)	135,913	36.4	38.0	11.7
<i>PD</i> * <i>LGD</i> _{After} (%)	136,830	0.38	0.14	0.71
Risk-weight (%)	136,629	63.1	57.4	37.5

This table summarizes the characteristics of credits and banks in the sample period from 2010Q2 and 2013Q3. The sample is restricted to term loan or revolving with at least two reporting banks. For credits, one observation is a credit-quarter. *Commitment size* is the total potential commitment of the credit across all banks. *Utilized* is the drawn dollar value and *Drawn* is the percentage of commitment utilized. *Participants* denotes the number of syndicate participants and *Reporting participants* is the number that report Basel II risk metrics. *Public* is a dummy equal to one if a borrower is public. *Age* is the number of quarters since origination. *New* is an indicator for the first quarter of a loan. The risk metrics are the average of the syndicate's reporting banks in the credit-quarter. For banks, the unit of observation is a bank-quarter. *Common credits* is the number of credits the bank participates in that have at least two reporting banks. *Participation rate* is the percentage of all credits in the credit sample that the bank participates in for the bank-quarter. *Foreign* is an indicator for a non-U.S. bank. Lastly, we present statistics for bank-credit-quarters. *Agent bank* is an indicator for agents in a credit syndicate. *Share of credit* is the percentage of a credit held by the bank. *PD* is the borrower probability of default. *LGD*_{Bef} and *LGD*_{After} is the loss given default before and after credit risk mitigants, respectively. *Risk weight* is the implied risk weight per dollar of EAD; it is calculated using *PD*, *LGD*_{After}, and the maturity of the loan.

The final panel of Table 1 summarizes characteristics that vary across banks but within credit-quarters (i.e., bank-credit-quarters). On average, 25% of observations are syndicate agents and reporting banks own 13% of credits. We consider three risk metrics in our analysis: PD, LGD, and the credit risk weight. We focus much of our analysis on PDs, as these are *borrower*-specific risk estimates that should not rely on the identity of the bank. In contrast, LGD and EAD can reflect bank-specific capabilities or experiences. Nevertheless, we report LGDs before and after credit enhancements, the expected percentage loss (the product of LGD and PD) and the implied risk weight of the loan.

2. Empirical Analysis

Our empirical analysis advances in three stages. First, we test for the presence of systematic bank-level differences in internally generated risk metrics. We find that there are significant differences across banks within credits that result in meaningful differences in risk weights. Second, we explore potential causes of these differences with an emphasis on the role of capital constraints. And third, we examine whether capital constrained banks' risk estimates are consistent with the prices they charge on loans.

2.1 Bank effects and risk metrics

Given the AIRB was designed to allow banks to use proprietary information to estimate risk, we expect risk estimates to differ across banks. However, systematic differences across banks are more difficult to explain: proprietary information should improve the *accuracy* of risk estimates rather than the *level* of risk.

We test whether there are significant bank effects by comparing risk estimates within credit syndicates. The deviation in PD, $\Delta PD_{i,j,t}$, for bank i in credit j at quarter t is:

$$\Delta PD_{i,j,t} = PD_{i,j,t} - \overline{PD_{j,t}},$$

where $\overline{PD_{j,t}}$ is the average PD of reporting banks. We repeat this process using the natural log of the risk metric, and denote these deviations as $\% \Delta$. The deviation from the syndicate average offers a straightforward demonstration of differences across banks, but we will also consider specifications in which we estimate banks' differences using credit-time fixed effects to account for potential correlations among regressors (Gormley and Matsa 2014).

Our objective is to statistically compare these differences across banks. We do so by regressing deviations on bank fixed effects, γ_i ,

$$\Delta PD_{i,j,t} = \gamma_i + \epsilon_{i,j,t}. \quad (1)$$

To test for the significance of these fixed effects, we conduct an F -test that the bank fixed effects are jointly equal.¹¹ This is analogous to a series of means difference tests across bank portfolios. An added benefit to the regression formulation is that we are able to allow for correlations in standard errors, such as repeated observations of the same credit or borrower. We present standard errors robust to heteroscedasticity and clustered two ways by borrower and by quarter, although the estimates are robust to a variety of alternative clustering specifications.

2.1.1 Results. Our results point to significant bank-level differences in internally generated risk metrics, Table 2. In Columns 1 and 2, we can reject

¹¹ F -statistics are adjusted to account for the degrees of freedom lost by estimating the mean for each credit-quarter.

Table 2
Regressions of risk metric deviations on bank fixed effects

Dep. var.:	Δ					% Δ	
	(1) <i>PD</i>	(2) <i>LGD</i> _{Bef}	(3) <i>PD</i> * <i>LGD</i> _{Bef}	(4) <i>PD</i> * <i>LGD</i> _{Aft}	(5) <i>RW</i> %	(6) <i>PD</i>	(7) <i>RW</i> %
Bank FEs:							
1	0.08***	3.85***	0.08***	0.09***	11.92***	0.14***	0.24***
2	0.48***	1.66***	0.12***	0.13***	14.76***	0.54***	0.28***
3	-0.32***	-4.38***	-0.15***	-0.15***	-17.82***	-0.35***	-0.40***
4	-0.06**	-0.35	-0.01	-0.01	0.67	0.00	0.04*
5	0.05	7.90***	0.07***	0.07***	10.35***	-0.02	0.26***
6	-0.04	-2.49***	-0.04***	-0.04***	-4.42***	0.06*	0.00
7	-0.17***	-2.59***	-0.08***	-0.08***	-10.34***	-0.18***	-0.23***
8	0.13***	-6.80***	-0.00	-0.00	-8.10***	-0.03	-0.23***
9	0.15**	-1.83**	-0.05***	-0.04***	-2.82**	0.16***	-0.04**
10	0.26***	-3.03***	0.03	0.03	4.23**	0.09***	0.03
11	0.19***	1.52*	0.07***	0.07***	2.70**	0.08**	0.06**
12	0.09***	0.16	0.04***	0.04***	4.18***	0.10***	0.08***
13	0.37***	-5.50***	0.09***	0.09***	2.18	0.40***	0.05
14	-0.41***	0.84**	-0.11***	-0.11***	-6.78***	-0.49***	-0.14***
15	0.09***	4.55***	0.04***	0.04***	8.98***	0.15***	0.20***
<i>F</i> -statistic	49.9	240.8	41.1	52.1	180.4	360.5	173.0
Observations	142,184	136,454	136,463	136,069	136,065	145,084	138,786
<i>R</i> -squared	0.04	0.28	0.05	0.05	0.22	0.14	0.23

This table regresses deviations in risk metrics on bank fixed effects. Deviations are calculated each quarter relative to the average risk metric in the credit syndicate. In Columns 1–5, the dependent variable is the level deviation; in Columns 6 and 7, it is deviation in logs. The sample consists of all bank-credit-quarters with more than one reporting bank from 2010Q2 to 2013Q3. *PD* is probability of default as a percentage. *LGD* is loss given default as a percentage. *LGD* and its dependent variables are reported before (*Bef*) and after (*Aft*) credit-risk mitigants. *RW*% is the percentage risk weight per dollar of EAD; it is calculated using *PD*, *LGD*_{Aft}, and the maturity of the loan. The *F*-stat tests the hypothesis that bank fixed effects are equal. Standard errors are clustered by bank-quarter but suppressed for brevity. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

the null hypothesis that bank fixed effects are equal for average deviations in PDs and LGDs before hedging (*LGD*_{Bef}) as the reported *F*-statistics are well in excess of relevant critical values (~ 2.1). We can also reject that bank effects in LGDs offset the effects in PDs; bank fixed effects are statistically different for the product of PD and LGD before hedging, Column 3, and after hedging, Column 4. These systematic differences carry over to the calculation of risk weights, Column 5. These findings are robust to a difference in log PDs specification, Column 6, and log risk weights, Column 7.

The presence of statistically significant bank fixed effects indicates that for a commonly held credit, internal risk metrics and risk weights are different across banks and that these differences are correlated with institutions to a degree that is not random. Moreover, the magnitude of the coefficients is economically meaningful. For example, the PD deviations in (1) range from -47 bps to as large as 53 bps, meaning that on average one bank reports PDs that are 30% lower than the bank-level average PD of 159 bps and another on average reports PDs that are 33% higher. The log regressions suggest that these differences are even larger, with the average percentage difference in PDs ranging from -49% to positive 54%. We will show that these magnitudes are important when calculating risk weights and capital ratios.

Figure 1 illustrates that the orientation of bank effects is robust to alternative empirical specifications. For both logs and levels, the direction of bias is fairly persistent as we move from the fundamental risk metrics like PDs to derived measures like risk weights. When we compare the value of fixed effects for PDs to those for risk weights (Figure 1A) twelve of the fifteen fall in the first and third quadrants indicating that they share the same sign. This is also true when we compare log risk metrics (Figure 1B) where thirteen of the fifteen bank effects share the same sign. In addition, the signs are consistent whether our deviations are in levels or logs (Figure 1C). These plots demonstrate that the ordering of bank effects, from high to low, is robust across risk measures.

If bank fixed effects are correlated with the risk metric averages, then the bank effects will be biased. To address this concern, we estimate regressions where the dependent variable is the risk metric rather than the deviation and bank effects are jointly estimated with credit time fixed effects (details are outlined in Internet Appendix Section IA.2). Additionally, we control for the role of the bank in a particular syndicate by including agent bank indicators and the share of the commitment held by each bank. Lastly, we consider a subsample of credits that we can match Loan Pricing Corporation's (LPC) Dealscan database in order to account for different roles of banks in the syndicate.

In these alternative specifications, we observe that bank fixed effects are positively correlated across empirical models, Figures 1D, 1E and 1F, which suggests the specification and controls do not explain our findings. More importantly, we find *F*-statistics that easily reject that bank fixed effects are equal (see Internet Appendix Table IA1).

2.2 Capital constraints and risk metrics

In this section, we will explore the source of risk metric differences that we identify above with an emphasis on the role of capital constraints. The Tier 1 ratio is a key regulatory ratio that reflects the value of Tier 1 capital to risk-weighted assets. The lower the Tier 1 capital ratio the closer a bank is to its capital constraint and the more scarce bank capital is within the institution.¹² As capital becomes scarce, there is an increased incentive to minimize the risk weight of investments which reduces the necessary regulatory capital held.¹³

The growing importance of risk estimates in regulation has been accompanied by their increased use in risk management, origination decisions, and performance evaluation. Specifically as it pertains to syndicated lending, the loan book typically receives a capital allocation and bankers seek to participate in loans that maximize performance within their capital budget. In this way,

¹² Despite capital ratios that are consistently in excess of regulatory constraints, evidence suggests capital constraints are binding as banks seek to maintain target capital buffers in excess of minimum requirements (see, e.g., Aiyar, Calomiris, and Wieladek 2014).

¹³ For example, to maintain a target Tier 1 capital ratio of 10% against a \$100 in assets, a bank must hold \$5 in capital if the risk weight is 50% and \$15 if the risk weight is 150%.

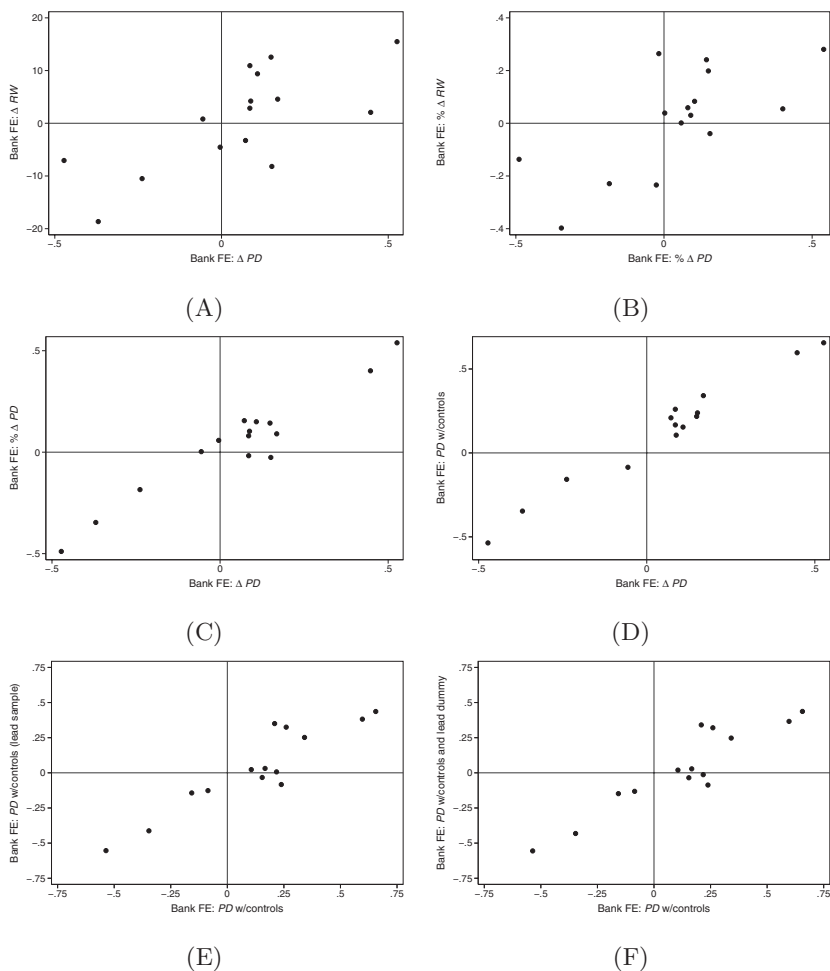


Figure 1
Comparison of bank fixed effects across risk metrics and models

This figure contains scatter plot comparisons of estimated bank fixed effects (FEs) across six pairs of regression models. Figure 1A compares FEs on level deviations in PDs in Table 2, Column 1, to FEs on level deviation in risk weights in Table 2, Column 5. Figure 1B compares FEs on log deviations in PDs in Table 2, Column 6, to FEs on log deviations in risk weights in Table 2, Column 7. Figure 1C compares FEs on level deviations in PDs in Table 2, Column 1, to FEs on log deviations in PDs in Table 2, Column 6. Figure 1D compares FEs on level deviations in PDs in Table 2, Column 1, to FEs on PDs in the presence of credit-quarter FEs and bank-credit controls in Internet Appendix Table IA1, panel A, Column 1. Figure 1E compares FEs on PDs in the presence of credit-quarter FEs and bank-credit controls in Internet Appendix Table IA1, panel A, Column 1, to FEs on PDs in the presence of credit-quarter FEs and bank-credit controls but is restricted to the sample matched to DealScan in Internet Appendix Table IA1, panel B, Column 1. Figure 1F compares FEs on PDs in the presence of credit-quarter FEs and bank-credit controls in Internet Appendix Table IA1, panel A, Column 1, to FEs on PDs in the presence of credit-quarter FEs, bank-credit controls, and a lead arranger dummy in the sample matched to DealScan in Internet Appendix Table IA1, panel C, Column 1.

bank-level capital constraints are transmitted to syndicated loan origination decisions and by extension risk assessments. When capital is scarce, there is a greater incentive on the part of bankers to burnish risk estimates, thereby reducing the capital required to be held against loans and improving the return on capital for the portfolio of loans.

If there are no costs of this behavior, we would expect all banks to report risk estimates near zero. However, internal controls and supervision impose constraints on this activity. Our hypothesis is that tighter capital constraints result in lower risk estimates, *ceteris paribus*. We test this hypothesis by estimating the relation between within-credit differences in risk estimates and measures of capital constraints. Our primary measure of capital constraints is the Tier 1 capital ratio, but we consider several related alternatives. If more constrained banks bias their risk estimates lower, we expect to find a positive relation between risk differences and the Tier 1 ratio.

A key concern when interpreting correlations between deviations in risk metrics and capital constraints is that the Tier 1 ratio is endogenous. There are two mechanisms that might attenuate the relation between risk metrics and capital ratios. The first is reverse causality. The direct effect of downward-biased PDs is to lower RWA and raise the Tier 1 ratio. Therefore, lower PD banks should have a higher Tier 1 ratio all else equal. A second form of simultaneity is learning. If banks suffer a loss and learn their portfolio is riskier, they should present a lower Tier 1 ratio and *increase* risk metrics. These two mechanisms bias our tests toward not finding a positive relation between risk metrics and bank capital.

Of greater concern are stories that would result in a positive correlation between risk metrics and bank capital. We present and investigate these stories in Section 2.3 following the main results.

2.2.1 Bank-level results. We begin our analysis by aggregating across credits to compare differences in risk metrics at the level of the loan portfolio. To generate portfolios we sum our within-credit differences in PDs and risk weights to the bank-quarter level, weighting by the utilized value of the loan to account for the greater importance of large loans in capital budgeting.¹⁴ Figure 2 plots weighted PD level deviations versus the Tier 1 capital ratio for each bank-quarter. Consistent with the capital constraint hypothesis, there is a positive correlation – better capitalized firms report higher PDs. The difference in weighted PD is as large as 150 bps across banks. We estimate this relation statistically using a pooled cross-sectional regression,

$$\Delta PD_{i,t} = \beta_0 Capital_{i,t} + \beta_1' \mathbf{BankControls}_{i,t} + \tau_t + \varepsilon_{i,t} \quad (2)$$

where the coefficient of interest is the relation between capital measures and risk metric deviations, β_0 . We condition on bank characteristics and time fixed

¹⁴ The utilized value is the drawn portion for revolving credit facilities and the outstanding principal for term loans.

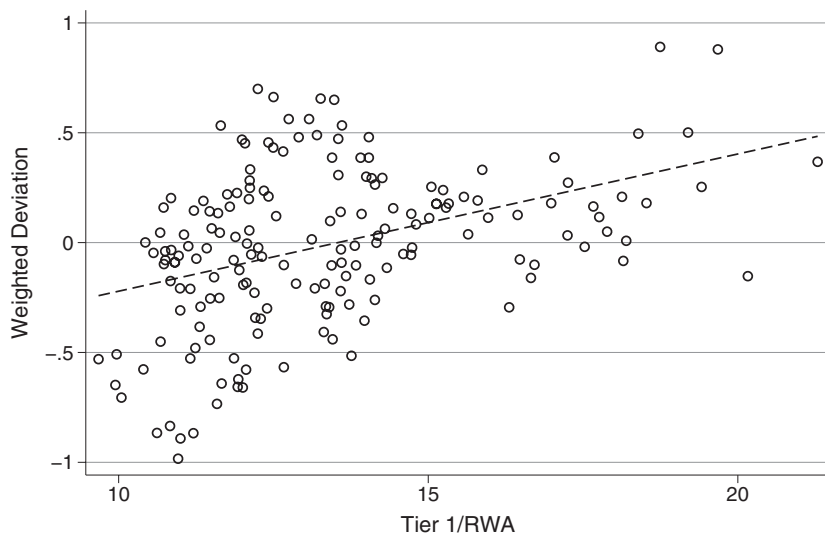


Figure 2

Bank portfolio PD deviations relative to the Tier 1 capital ratio

This figure plots the weighted sum of PD deviations by bank-quarter versus the Tier 1 capital ratio. The average is weighted by the share of utilized funds for that bank-quarter.

effects, τ_i . Because we have aggregated to bank portfolios, we cannot cluster by borrower. Clustering by bank is also questionable given the small number of resultant clusters relative to the number of observations, therefore we calculate heteroscedasticity autocorrelation consistent (HAC) standard errors using a bandwidth of six quarters, consistent with the median length of a loan.¹⁵ While the results are robust to various standard error choices, we should caveat this analysis by noting the natural limitations of statistical inference when confronted with a short time-series and/or a narrow panel.

In addition to the Tier 1 capital ratio, we calculate an alternative measure of banks' regulatory constraints which is meant to capture the distance of a bank from its "target" Tier 1 ratio where the target is a function of bank characteristics and aggregate conditions, *Tier 1 gap*.¹⁶ This is formed by taking the residuals from a regression of the Tier 1 capital ratio on log assets, ROE, leverage, quarter fixed effects, and a foreign bank dummy. The residuals are estimated quarterly for every bank in our sample for the full period 2009Q3–2013Q3 regardless of when they enter the SNC.

Table 3 shows that banks with a lower Tier 1 ratio, Column 1, or a lower Tier 1 gap, Column 2, report lower PDs for a common set of credits. If Tier 1

¹⁵ In other words, our results are robust to arbitrary serial correlation with a six-quarter lag.

¹⁶ See Berger et al. (2008) for evidence that banks target capital ratios and that these targets are related to observable characteristics, in particular bank size.

Table 3
Bank-Level: Regression of portfolio deviations on capital measures

Δ :	(1) <i>PD</i>	(2) <i>PD</i>	(3) <i>RW%</i>	(4) <i>RW%</i>	(5) <i>PD</i>	(6) <i>PD</i>	(7) <i>PD</i>	(8) <i>PD</i>
Tier 1/RWA	0.079*** (0.019)		1.065*** (0.303)				0.054*** (0.014)	
Tier 1 gap		0.081*** (0.020)		1.229*** (0.271)				0.051*** (0.015)
<i>tier 1/assets</i>					1.366*** (0.327)			
<i>assets/rwa</i>					1.157*** (0.270)			
Tier 1/RWA _{<i>t</i>-1}						0.076*** (0.020)		
<i>assets</i>	-0.015 (0.040)	-0.089** (0.042)	-1.587 (1.768)	-2.631 (1.777)	-0.006 (0.040)	-0.013 (0.042)	0.099 (0.236)	0.026 (0.230)
Foreign	-0.252** (0.113)	-0.128 (0.095)	-6.024* (3.603)	-4.363 (3.492)	-0.101 (0.091)	-0.249** (0.117)	-1.237*** (0.456)	-1.100** (0.447)
Year FEs	+	+	+	+	+	+	+	+
Bank FEs							+	+
Observations	174	174	174	174	174	174	174	174
<i>R</i> -squared	0.23	0.24	0.08	0.10	0.26	0.21	0.08	0.15

This table regresses the sum of level deviations in risk metrics on measures of capital adequacy. Deviations are calculated each quarter relative to the average risk metric in the syndicate. Deviations are weighted by the drawn dollar value for a given bank-quarter and then summed to the bank level. The sample consists of bank-quarters from 2010Q1 to 2013Q3. *PD* is probability of default as a percentage. *RW%* is the risk weight per dollar of EAD as a percentage; it is calculated using *PD*, *LGD_{Apr}*, and the maturity of the loan. *Tier 1/RWA* is the most recent reported Tier 1 capital ratio; *Tier 1 gap*, is the estimated deviation from an expected Tier 1 ratio; *Tier 1/assets* is the log of Tier 1 capital to assets; *Assets/rwa* is log of assets to risk-weighted assets; the *log(Assets)* is the log of total assets; and *Foreign* is a dummy for non-U.S. banks. All regressions include time fixed effects; the final two columns include bank fixed effects. Standard errors reported in parentheses are HAC within a six-quarter lag. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

gap decreases by 10% the downward bias in weighted-PD increases by 79 bps. Both capital constraint measures explain a sizable portion of the cross-sectional variation in PD deviations. In univariate regressions, Tier 1 capital and Tier 1 gap generate adjusted *R*-squared of 16% and 20%, respectively.¹⁷

The magnitude of a 79 bps change has a significant impact on the capital allocated to the corporate loan portfolio. Figure 3 illustrates the sensitivity of risk weights to PD.¹⁸ A 75-bps change from 150 bps PD to 75 bps decreases RWA from 90% to 70% reducing the necessary capital for corporate loans by over 20%. When we estimate the relation between Tier 1 capital and risk weight percentages, Columns 3 and 4, we find magnitudes that are attenuated. A 10% decrease in Tier 1 implies an 11%-12% lower risk weight. The attenuation is expected, as risk weights are ultimately the function of additional choices, such as hedges and loss assumptions. In the remainder of the paper we focus on PDs as this is the risk metric that is most comparable across syndicate-member banks.

¹⁷ The interpretation of log differences is less clear at the portfolio level; however, we repeat the analysis with weighted average log differences in Internet Appendix Table IA2. The results are qualitatively similar.

¹⁸ Maturity of 3 years, LGD of 35% and an EAD of 100%.

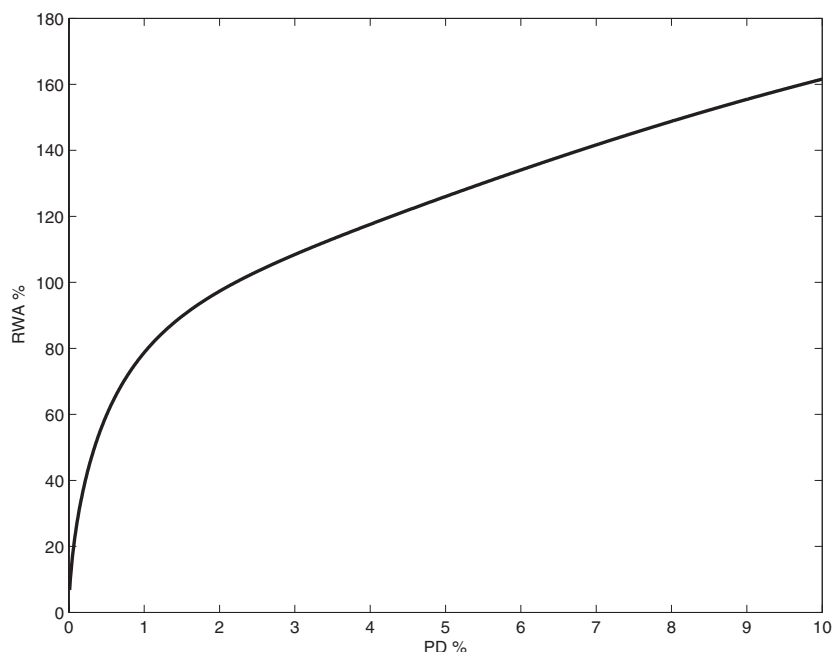


Figure 3

Risk-weighted assets as a function of the probability of default

This figure plots the RWA as a function of PD under the Basel II AIRB. This relation is based on average sample values for maturity (3 years) and LGD (35%). We also assume an EAD of 100%.

To better understand the source of covariation between PD and capital ratios, we decompose the Tier 1 ratio into two distinct terms. The Tier 1 ratio is the product of Tier 1 capital to assets and assets to risk-weighted assets. Taking logs of this product we can write the Tier 1 ratio in two parts: one reflecting the capital of the firm and the other the riskiness of the balance sheet. In Column 5, we can see that the relation with PD differences is increasing and of similar magnitude for both terms, hence variation in both the capital position and the riskiness of the firm contribute to the positive coefficient on the Tier 1 ratio. Banks with less capital report lower PDs as do banks with more risky assets.

Our analysis focuses on contemporary measures of capital constraint as we assume firms have well-formed expectations about their near-term capital levels. Nevertheless, in Column 6 we verify that the lagged Tier 1 ratio produces similar results.

The Tier 1 capital ratio might reflect a bank's private views on risk. A bank with a low Tier 1 ratio is a bank with a higher risk-weighted leverage. Assuming this is a bank's preferred capital structure, higher leverage is consistent with a greater tolerance for risk or a lower perception of overall risk. Either of these mechanisms could also influence internally generated PDs. To account for persistent bank characteristics, we include bank fixed effects in Columns

Table 4
Credit-level regression of PD on the tier 1 ratio

Dep. var:	(1) <i>PD</i>	(2) <i>PD</i>	(3) <i>PD</i>	(4) <i>log(PD)</i>	(5) <i>log(PD)</i>
Tier 1/RWA	0.084*** (0.017)	0.029** (0.011)	0.031** (0.012)	0.062*** (0.012)	0.031** (0.012)
Agent	−0.127*** (0.029)	−0.121*** (0.029)	−0.144** (0.049)	−0.052* (0.025)	0.717*** (0.155)
Share	−0.869*** (0.267)	−0.626** (0.259)	−1.603*** (0.528)	−0.158*** (0.043)	−0.872*** (0.081)
Bank controls	+	+	+	+	+
Credit-quarter FEs	+	+	+	+	+
Bank FEs		+	+		+
Weighted			+		+
Observations	142,113	142,113	95,857	145,084	98,403
<i>R</i> -squared	0.62	0.63	0.69	0.74	0.80

This table regresses PD on the Tier 1 ratio conditional on credit-date fixed effects. The sample consists of bank-quarters from 2010Q1 to 2013Q3. *PD* is probability of default as a percentage, *log(PD)* is the log of PD. *Tier 1/RWA* is the most recent reported Tier 1 capital ratio. Bank-credit controls include an agent bank indicator (*Agent*), and the share of the credit a bank owns (*Share*). Suppressed bank controls include log of bank assets and a foreign bank indicator. All specifications include credit-quarter fixed effects; Columns 2, 3, and 5 include bank fixed effects. Columns 3 and 5 are weighted by the drawn size of the credit relative to the bank's observed loan portfolio. Standard errors reported in parentheses are clustered two ways by borrower and by date. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

7 and 8. Regressions with bank fixed effects are more robust to alternative interpretations, but bank fixed effects absorb some cross-sectional variation in banks that might be attributable to capital constraints. We continue to find that the coefficients on capital constraints are positive and statistically significant at the 1% level, but the magnitude of the capital coefficients is 10%–20% smaller than what we observe in specifications without fixed effects.

2.2.2 Credit-level results. We continue our investigation using a credit-level analysis. To that end, we jointly estimate the relation between PD and capital in the presence of credit-date fixed effect,

$$PD_{i,j,t} = \beta_0 Capital_{i,t} + \beta'_1 \mathbf{BankControls}_{i,t} + \beta_2 Agent_{i,j,t} + \beta_3 Share_{i,j,t} + \mu_{j,t} + \gamma_i + \varepsilon_{i,j,t} \tag{3}$$

where the coefficient of interest is β_0 . We include credit-date fixed effects, $\mu_{j,t}$, to focus on within credit differences in PDs. We condition on bank characteristics and in some specifications we include bank fixed effects, γ_i . Every specification includes bank-credit controls: a dummy indicating whether the bank is the agent for the credit facility, *Agent*, and the share of the credit the bank owns, *Share*. In a subset of specifications, we weight by the share of the loan in a bank's portfolio to account for the greater importance of larger credits for a bank's capital decisions. Standard errors are robust to heteroskedasticity and clustered two ways by borrower and by quarter to account for repeated borrowers and common unobserved shocks at a point in time.

The credit-level analysis summarized in Table 4 broadly confirms our bank-level finding that differences in PDs are positively correlated with measures of

capital constraints. We find that the coefficient on the Tier 1 ratio is 0.082 and statistically significant when credits are equally weighted, Column 1. When we include bank fixed effects, Column 2, the magnitude of the coefficient is smaller, 0.045, but remains statistically significant at the 1% level. When we weight by the size of the loan relative to the bank's portfolio (Column 3), we lose approximately 46,500 credit-quarters which are undrawn credits; however, the coefficient increases slightly to 0.058 which suggests PDs on larger, drawn credits are more sensitive to capital constraints. Columns 4 and 5 demonstrate similar outcomes using log PDs on the left-hand side rather than the level of PDs. The magnitudes are also consistent with the bank results in Table 3. Lastly, we consider the Tier 1 gap measure of capital constraints in Internet Appendix Table IA3 and draw similar conclusions.

2.3 Robustness tests

Bank fixed effects account for persistent differences across banks, but risk attitudes or perceptions might change over time within a bank. If so, PDs and capital ratios will move together. To account for this possibility, we consider two time-varying proxies for the risk posture of banks. Both presume that banks with a more optimistic perception of risk will hold riskier portfolios on average. The first measure is the riskiness of the commonly held loan portfolio, calculated as the sum of mean PDs reported by other banks in a loan syndicate weighted by the value drawn.¹⁹ The second is the overall riskiness of the firm as proxied for by the ratio of total risk-weighted assets to assets.

The results are summarized in Table 5. We focus these additional tests on the regression specification with bank fixed effects and weighted by loan size (i.e., Table 4, Column 3). The risk proxies do not attenuate the coefficient on capital for either the Tier 1 ratio, Columns 1 and 2. In fact, the coefficient on both capital measures is greater than the equivalent specification in Table 4. Neither risk proxy is statistically significant.

If banks selectively choose their participation in loans, our results might be biased. Capital constrained banks could be "pickier," choosing to concentrate their holdings in credits that they believe earn the highest spread relative to the risk they bear. Hence, we observe only those credits where their private information suggests they have a differentially low view on the risk of the credit. Such behavior would result in banks holding credits where their internal estimates are low relative to other banks and avoiding credits where their estimates are higher. If this form of selectivity is a persistent bank trait, it will be captured by bank fixed effects. To account for this explanation, we include the participation rate of banks in the universe of credits. If banks are only selecting the relatively less risky credits from an unconditional distribution that resembles their peers, then their participation rate should be lower.

¹⁹ Formally, the portfolio risk for bank i at time t is $\overline{PD}_{i,t} = \sum_j \overline{PD}_{j,t} \frac{Utilized_{i,j,t}}{\sum_j Utilized_{i,j,t}}$, where $\overline{PD}_{j,t}$ is the average reported riskiness of a loan excluding bank i .

Table 5
Credit-level regression of PD on the Tier 1 ratio, risk proxies, and activity

Dep. var:	(1) <i>PD</i>	(2) <i>PD</i>	(3) <i>PD</i>	(4) <i>PD</i>	(5) <i>PD</i>	(6) <i>log(PD)</i>	(7) <i>log(PD)</i>
Tier 1/RWA	0.087*** (0.017)	0.114*** (0.019)	0.068*** (0.016)	0.123*** (0.019)	0.054*** (0.016)	0.096*** (0.014)	0.061*** (0.016)
Portfolio risk	-0.089 (0.060)			-0.003 (0.077)	0.219** (0.096)	-0.121* (0.059)	0.064* (0.033)
RWA/Assets		0.008*** (0.002)		0.017*** (0.002)	0.011 (0.010)	0.015*** (0.002)	0.016** (0.006)
Mkt. participation			-1.130*** (0.116)	-1.785*** (0.186)	1.055 (1.269)	-1.997*** (0.286)	3.232*** (0.923)
Controls	+	+	+	+	+	+	+
Credit-quarter FEs	+	+	+	+	+	+	+
Bank FEs					+		+
Weighted					+		+
Observations	142,113	142,113	142,113	142,113	95,857	145,084	98,403
R-squared	0.62	0.62	0.62	0.62	0.69	0.76	0.80

This table regresses PD on the Tier 1 ratio in the presence of proxies for potentially omitted variables. The sample consists of bank-quarters from 2010Q1 to 2013Q3. *PD* is the probability of default as a percentage, *log(PD)* is the log of PD. *Tier 1/RWA* is the most recent reported Tier 1 capital ratio. Controls include log of bank assets, a foreign bank indicator, an agent bank indicator, and the share of the credit a bank owns. *Portfolio risk* is the weighted average PD of credits in the bank's portfolio based on the reporting of other banks in the credit. *RWA/assets* is the ratio of risk-weighted assets to total book assets. *Mkt. participation* is the percentage of outstanding credits the bank participated in that quarter. All specifications include credit-quarter fixed effects; Columns 5 and 7 include bank fixed effects and are weighted by the size of the credit relative to the banks observed loan portfolio. Standard errors reported in parentheses are clustered two ways by borrower and by date. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

We find that banks with a lower participation rate also tend to report lower credit risks, Column 3, but that the participation rate is neither statistically significant nor does it meaningfully diminish the correlation with capital ratios. We include all of these proxies in Columns 4 and find similar coefficients of around 0.6 on capital constraints, albeit at slightly lower levels of statistical significance. We obtain similar results when we exclude bank fixed effects and equal-weight credits, Column 4, or when consider log PDs rather than levels, Columns 5 and 6. Lastly, we confirm these findings using the Tier 1 gap measure of capital constraints in Internet Appendix Table IA4.

We repeat our analysis using alternative sample groups to address several additional identification concerns. One such concern is that the relation between risk metrics and capital is driven by differences in supervisory regimes. While bank fixed effects should control for bank-specific factors, like the identity of the supervisory agency, it may be that some supervisory regimes drive this behavior. To address this concern we divide the sample into U.S. and non-U.S. lenders.

Table 6 Columns 1 and 2 contain the Tier 1 ratio coefficients for foreign and domestic banks, respectively. We consider four specifications for each subsample. Panel A considers the basic empirical specification with credit-date fixed effects and bank controls, and panel B includes empirical specifications that are weighted by the drawn value of the loan and that contain bank fixed effects and the proxies for risk attitudes and bank activity (e.g., Table 5,

Table 6
Credit-level subsample regressions of PD on the Tier 1 ratio

A	(1)	(2)	(3)	(4)	(5)	(6)
Sample:	Non-U.S.	U.S.	DealScan	Lead	Nonleads	Nonagents
Dep. var.: PD						
Tier 1/RWA	0.093*** (0.016)	0.076*** (0.023)	0.066*** (0.016)	0.086*** (0.028)	0.049*** (0.015)	0.097*** (0.015)
Observations	23,625	100,981	64,302	16,683	34,620	46,761
R-squared	0.69	0.60	0.60	0.63	0.57	0.64
Dep. var.: log(PD)						
Tier 1/RWA	0.068*** (0.009)	0.071** (0.024)	0.056*** (0.014)	0.056* (0.028)	0.055*** (0.013)	0.067*** (0.009)
Observations	23,837	103,192	64,853	16,820	34,890	47,843
R-squared	0.86	0.70	0.72	0.71	0.73	0.76
B	(1)	(2)	(3)	(4)	(5)	(6)
Sample:	Non-U.S.	U.S.	DealScan	Lead	Nonleads	Nonagents
Dep. var.: PD						
Tier 1/RWA	0.077** (0.029)	0.092 (0.059)	0.035 (0.024)	0.151** (0.058)	0.021 (0.030)	0.056** (0.019)
Observations	15,139	66,769	39,073	10,156	18,923	31,766
R-squared	0.73	0.65	0.68	0.71	0.66	0.70
Dep. var.: log(PD)						
Tier 1/RWA	0.064*** (0.015)	0.199*** (0.053)	0.048*** (0.016)	0.142*** (0.041)	0.054*** (0.013)	0.064*** (0.019)
Observations	15,322	68,602	39,541	10,263	19,161	32,612
R-squared	0.83	0.76	0.79	0.82	0.80	0.81

This table regresses PD on the Tier 1 ratio for six subsamples: (1) the sample of foreign banks, (2) the sample of U.S. banks, (3) the sample of credits we match to DealScan, (4) the sample of lead arrangers identified in the DealScan data, (5) the sample of nonlead arrangers in the DealScan matched data, and (6) nonagent banks in the credits not matched to DealScan. Results are presented for two dependent variables: *PD* is the probability of default as a percentage, *log(PD)* is the log of PD. *Tier 1/RWA* is the most recent reported Tier 1 capital ratio. Panels A includes credit-quarter fixed effects and bank controls, where bank controls are log of bank assets, a foreign bank indicator, an agent bank indicator, and the share of the credit a bank owns. Panel B also includes bank fixed effects, proxies for omitted variables used in Table 5, and regressions are weighted by the size of the credit relative to the banks observed loan portfolio. Standard errors reported in parentheses are clustered two ways by borrower and by date. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Column 5). Within each panel we present results for the level of the dependent variable as well as the log.

The coefficient on the Tier 1 capital ratio is consistent with the full sample results in Tables 4 and 5. In all eight specifications there is a positive correlation between PDs and the Tier 1 ratio. In seven of these specifications, the coefficient is statistically different from zero. The one exception is that the coefficient for U.S. banks in the presence of bank fixed effects and other controls has a *t*-stat of 1.55; nevertheless, the sign and magnitude of the coefficient is consistent with the other results. While there are level differences between foreign and U.S. banks, there is no consistent pattern that suggests a robust statistical difference between the two. These findings demonstrate our results are not unique to a particular regulatory regime, but they also underscore that no single bank can explain the results as the two samples are mutually exclusive.

Another concern is whether the role of the bank in the syndicate matters. Thus far we have controlled for agent bank status and the participation of each bank in the syndicate (SNC identifies only the agent bank, i.e., the primary lead). However, other lead arrangers might have better access to borrower information. If those roles are correlated with capital ratios and there are trends in risk metrics over time due to the flow of information, then the result could be a spurious correlation between risk metrics and capital ratios. While it is not obvious why information would have a directional effect on PDs, we investigate this issue closely. To further address the role of banks in syndicates, we match our sample of credits to Dealscan in order to identify all arrangers in a credit.

Before conditioning on lead arranger status, we establish the relation between credit constraints and PDs within this matched sample, Table 6 Column 3. In all four specifications, the coefficient is positive but attenuated relative to the full sample results in Tables 4 and 5. Nevertheless the coefficient is statistically significant at the 1% level in three of the four specifications and marginally so in the fourth (t -stat of 1.45). The statistical attenuation is likely the result of the smaller sample size after matching to DealScan, but is also affected by the change in sample composition. The typical credit matched to the DealScan data is larger, less risky, and more likely to be public, than an unmatched credit.²⁰ In the next section, we explore the role of these characteristics and we find that these sample characteristics tend to attenuate the relation between PDs and capital constraints.

When we condition on lead arranger status, the relation between credit constraints and risk metrics remains strong, Column 4, particularly in panel B, where the magnitudes are more than twice those in Table 5. Hence, the relation is robust even within the set of lenders that are best informed about the borrower. Column 5 uses a sample restricted to nonlead arrangers. The coefficient on the Tier 1 capital ratio is positive in all four specifications and statistically significant in three of the four. The specification that is statistically indistinguishable from zero is the same as that in Column 3: PD levels as the dependent variable with bank fixed effects, proxies for omitted variables, and weighted by size.

Given the attenuation observed in Column 3, we also consider a “nonleads” sample in the unmatched sample of credits by excluding agent banks, Column 6. Agent banks are typically the primary lead, so excluding them is analogous to the exclusion of lead arrangers in Column 5, but in a sample of credits for which we believe the incentive to tilt risk metrics is stronger. Note that we cannot conduct the analysis with agent banks as there is only one agent per credit. The coefficients for the nonagent sample closely resemble the full sample results and are statistically significant and positive in each specification. In sum, the

²⁰ Commitment size is \$576m versus \$450m unmatched; PDs are 1.35% versus 2.3% for unmatched; and 50% of matched credits are to public borrowers versus 38.5% for unmatched credits.

role of capital constraints on risk metric deviations is robust within types of lender roles.

A final set of concerns is that banks update their risk metrics at different frequencies. If update frequency varies across banks, then some metrics may be stale, whereas others are current. As with lead status, if this is correlated with capital ratios then our finding that more capital constrained banks report lower risk metrics could be spurious.

In our sample of credit-bank-date quarters, banks change their PDs 34% of the time or 1.4 times a year. LGDs change at a slightly lower frequency of 1.2 times a year. There is significant cross-sectional heterogeneity in the frequency with which banks vary PDs with some banks varying them as little as a 0.5 times a year and others 3 times a year. Such bank traits should be captured by fixed effects, but this tendency could be associated with bank role or capital. Indeed, we find that unconditionally agent banks are 5% more likely to change their PD relative to nonagents. In addition, capital is negatively correlated with the likelihood of changing PD, albeit not conditional on bank fixed effects.

To rule out concerns prompted by these correlations, we consider two alternative specifications. First, we condition on the set of observations where a PD has been changed, thereby eliminating the risk of stale PDs influencing our results. Second, we condition on the set of credits that are less than 6 months old, reducing the time for PDs to become stale.

The findings related to PD changes are summarized in Table 7. When we condition on changes, Columns 1 and 2, the sample size is roughly one-fifth the size relative to the full sample in both specifications: the base specification in panel A and the specification that includes bank fixed effects in panel B. The coefficient on the Tier 1 ratio is positive in both panel A and panel B as well as when we consider the level of PDs, Column 1, and log of PDs, Column 2, which is consistent with prior results. In addition, the difference from zero is statistically significant in three of the four specifications; the exception is the log PD specification that excludes bank fixed effects where the t -stat is 1.48, which corresponds to a p -value of 14%.

The results that condition on changes suggest that even within the set of recently updated PDs, there is a positive relation between capital constraints and risk metrics within credits. A similar subsample is outlined in Columns 3 and 4 where we consider new credits that are less than 6 months old (the first two quarters a credit is in the sample). This criterion significantly reduces the sample size, but in this case the sample captures the initial risk metrics for the original syndicate participants. In all four empirical specifications, Columns 3 and 4 for panels A and B, the coefficient on the Tier 1 ratio is positive. The coefficients are statistically different than zero in panel A and consistent in magnitude with prior findings. However, in panel B the statistical significance is attenuated. For levels, Column 3 the coefficient's p -value is 11%. Note that while these specifications are not statistically differentiated from zero, they are

Table 7
Credit-level PD changes and regressions of PD on the Tier 1 ratio

A Dep. var.:	Changes		< 6 months		Full sample	
	(1) <i>PD</i>	(2) <i>log(PD)</i>	(3) <i>PD</i>	(4) <i>log(PD)</i>	(5) <i>PD</i>	(6) <i>log(PD)</i>
Tier 1/RWA	0.079*** (0.024)	0.043 (0.029)	0.079*** (0.016)	0.058*** (0.012)	0.083*** (0.015)	0.057*** (0.011)
Tier 1/RWA * Year 2					−0.004 (0.009)	0.006 (0.006)
Tier 1/RWA * Year 3					0.014 (0.015)	0.033*** (0.011)
Tier 1/RWA * Year 3+					0.005 (0.018)	−0.005 (0.008)
Observations	24,153	24,654	24,746	25,026	142,104	145,084
R-squared	0.56	0.65	0.62	0.75	0.62	0.74

B Dep. var.:	Changes		< 6 months		Full sample	
	(1) <i>PD</i>	(2) <i>log(PD)</i>	(3) <i>PD</i>	(4) <i>log(PD)</i>	(5) <i>PD</i>	(6) <i>log(PD)</i>
Tier 1/RWA	0.168** (0.059)	0.072* (0.034)	0.069 (0.043)	0.028 (0.026)	0.031 (0.018)	0.048** (0.016)
Tier 1/RWA * Year 2					0.044** (0.015)	0.023*** (0.008)
Tier 1/RWA * Year 3					0.046* (0.026)	0.040*** (0.013)
Tier 1/RWA * Year 3+					0.014 (0.026)	0.010 (0.013)
Observations	16,787	17,199	14,781	15,011	95,856	98,403
R-squared	0.66	0.77	0.69	0.80	0.69	0.80

This table tests the sensitivity of the capital constraint coefficient to the age of the risk metric or the age of the credit. (1) and (2) restrict the sample of observations to bank-credit-quarters in which the bank's PD estimate changed from the prior quarter; (3) and (4) restrict to loans that are less than 6 months old; and, (5) and (6) use the full sample but include interactions with the age of the loan. Results are presented for two dependent variables: *PD* is the probability of default as a percentage, *log(PD)* is the log of PD. *Tier 1/RWA* is the most recent reported Tier 1 capital ratio. Panel A includes credit-quarter fixed effects and bank controls, where bank controls are log of bank assets, a foreign bank indicator, an agent bank indicator, and the share of the credit a bank owns. Panel B also includes bank fixed effects, proxies for omitted variables used in Table 5, and regressions are weighted by the size of the credit relative to the banks observed loan portfolio. Standard errors reported in parentheses are clustered two ways by borrower and by date. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

also not dissimilar from other coefficient estimates given the observed standard errors.

We further test the importance of the age of the credit by interacting the capital constraint measure with three dummies that reflect the age of the credit: a dummy for credits that are in their second year of origination, a dummy for credits in their third year of origination, and a dummy for credits older than 3 years. The uninteracted coefficient on Tier 1 capital captures the relation between capital constraints and PDs in the first year. The interaction terms can be interpreted as the difference in the capital constraint relation relative to the first year of origination.

If stale PDs are the cause of the within credit deviations and their correlation with capital constraints, then the relation in the first year should be lower than later years and indistinguishable from zero. In addition, we would expect the

interactions term coefficients to be positive and increasing over time as credits get older. The results in Column 5 and 6 fail to provide either of these patterns consistently. In panel A, the relation between capital constraints and PDs in the first year is statistically significant and of an economic magnitude consistent with the full-sample results. Moreover, there is no clear pattern that suggests the relation between capital and PD deviations is increasing over time. When we include the bank fixed effects and weight the regressions in panel B, we find positive coefficients in both Columns 5 and 6, and statistically significant results at the 5% level in the log specification in Column 6. For credits that are in their second or third year the relation with capital constraints is larger, however in the three plus category the coefficient is smaller and there is no statistical difference from the relation observed in the first year. Three of four specifications show a relation within the first year of a credit and there is no consistent pattern that suggests that older credits exhibit a greater sensitivity of PDs to capital constraints.

The analysis in Tables 6 and 7 produce similar coefficients as our primary findings. While there are idiosyncratic estimates with attenuated statistical significance, there does not appear to be a consistent pattern of dissimilar results. Also it does not appear that syndicate role, the age of the credit or asynchronous updating of risk metrics can explain the correlation of deviations within credits with capital.

2.4 Credit-level heterogeneity

Having ruled out several alternative explanations for our findings, in what follows we ascertain the role of capital incentives by investigating whether the sensitivity to capital is greater for particular types of credits. We regress PD on an interaction between a capital measure and a credit characteristic term, $X_{j,t}$,

$$PD_{i,j,t} = \beta_{0,X}(Capital_{i,t} * X_{j,t}) + \beta_2 Agent_{i,j,t} + \beta_3 Share_{i,j,t} + \mu_{j,t} + \gamma_{i,t} + \varepsilon_{i,j,t}. \quad (4)$$

Regressions include credit-time, $\mu_{j,t}$, and bank-time, $\gamma_{i,t}$, fixed effects which make the uninteracted capital term redundant. The coefficient of interest in Equation (4) is the interaction term, $\beta_{0,X}$ which summarizes the sensitivity of the relation between capital and PD differences to credit characteristics. Like in Equation (3), we include bank-credit controls. Because we wish to understand which types of credits correlate with capital, we do not weight these regressions which would tilt the results toward larger credits.²¹ Standard errors are robust to heteroscedasticity and are clustered two ways by borrower and by quarter.

²¹ This is in contrast to the earlier section where we cared about differences in large credits more because large credits contribute more risk to the bank.

Table 8
Credit-level regression of PD on the Tier 1 ratio interactions

Dep. var.:	(1) <i>PD</i>	(2) <i>PD</i>	(3) <i>PD</i>	(4) <i>PD</i>	(5) <i>PD</i>	(6) <i>PD</i>	(7) <i>log(PD)</i>	(8) <i>log(PD)</i>
T1*log(Drawn amt.)	0.023*** (0.006)	0.007 (0.006)			0.002 (0.006)	0.002 (0.007)	0.005 (0.003)	0.004 (0.003)
T1*Drawn		0.077*** (0.022)			0.033** (0.014)		0.042*** (0.009)	
T1*log(Mean PD)			0.055*** (0.017)		0.043** (0.017)	0.045* (0.023)	0.013*** (0.003)	0.013*** (0.004)
T1*Public				-0.058*** (0.019)	-0.034** (0.016)	-0.045* (0.021)	0.006 (0.007)	0.002 (0.008)
Sample	Full	Full	Full	Full	Full	Drawn	Full	Drawn
Observations	142,104	142,104	142,104	142,104	142,104	103,216	145,084	105,835
R-squared	0.64	0.64	0.62	0.62	0.62	0.60	0.80	0.76

This table regresses PD on the Tier 1 capital interactions with credit characteristics conditional on credit-date and bank-date fixed effects. Results are presented for two dependent variables: *PD* is the probability of default as a percentage, *log(PD)* is the log of PD. The sample is restricted to only drawn credits in Columns 6 and 8. *log(Drawn amt.)* is the log of the drawn amount attributable to the bank. *Drawn* is a dummy equal to one if the credit has an outstanding balance. *log(Mean PD)* is the log of the average PD of the credit. *Portfolio share* is the credits current utilized value for a bank scaled by the lending bank's total drawn portfolio. *Public* is a dummy equal to one for public firms. Each specification includes the requisite uninteracted terms, bank-date fixed effects, credit-date fixed effects, an agent bank indicator, and the share of the credit a bank owns. Standard errors reported in parentheses are clustered two ways by borrower and by date. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 8 summarizes our findings where we consider the Tier 1 ratio as our measure of capital constraints. The sensitivity of PD deviations to capital is positively related to several characteristics that increase the potential benefit of lower risk estimates to banks. The first characteristic is the size of the drawn portion of a credit. The larger the loan the greater exposure at default which increases the benefit of reporting lower risk. In Column 1, we interact capital measures with the log of each bank's exposure to the credit as captured by the utilized value. The correlation between capital and size is positive; however, drawn status captures not only the size of the credit but whether or not it has been drawn down if it is a revolver. To tease out the importance of drawn status versus size, we include in Column 2 an interaction with a dummy variable indicating whether the credit is drawn. The sensitivity of PDs to capital is significantly higher for drawn credits versus undrawn credits, a finding consistent with the fact that there is little benefit to a lower PD for undrawn credits. However, we find that larger credits, conditional on drawn status, are not more sensitive to capital constraints at conventional significance levels.²²

Next, we consider the riskiness of the loan. The impact of a change in PD on RWA declines as credits get riskier, see Figure 3; therefore, the higher the level of PD the greater a change is necessary to achieve a similar impact on RWA. Hence we should expect a greater sensitivity of PDs to capital the riskier the credit. We test this hypothesis by interacting capital measures with the log of

²² This is also true when we consider log PD as the dependent variable.

the average PD of the credit. We find riskier loans are more sensitive to capital, Column 2. The interaction is statistically significant at the 1% level.

Last, we consider the degree to which firm opacity correlates with the biasing behavior. Public companies produce readily available reports on financial conditions and as a result there is more “hard”, verifiable information available to all market participants. Consequently, banks have less discretion in their estimation of PDs for these firms. In contrast, for private firms banks must rely on private information when they formulate PDs, allowing for more maneuvering in the construction of risk estimates. Hence, we should expect that capital will be less sensitive to PDs for public, less opaque firms.

Indeed, when we estimate capital interacted with a public borrower dummy, we find that the interaction is negatively related to PD deviations which suggests that the relation between PDs and capital is smaller for credits to public borrowers, Column 4. The finding is statistically significant at the 1% level. For reference, if we do not include bank-time fixed effects and we estimate the coefficient on the Tier 1 ratio uninteracted, then the Tier 1 coefficient is 0.85 (significant at 1% level). The sum of the interacted (-0.58) and uninteracted coefficients is positive, but much closer to zero. The net effect is consistent with less discretion by banks in the formulation of PDs for public firms.

Public status is likely to be correlated with characteristics like size and risk. So in Columns 5 and 6 we consider these interaction terms jointly. Column 5 considers the full sample and draws similar conclusions as the single trait results: capital levels are more positively correlated with drawn, risky, and private credits. Each of these interaction terms is statistically significant at the 5% level. Column 6 conditions on the drawn status of the credit and also finds that PDs for risky and private credits are more sensitive to capital levels, albeit at the 10% significance level. We repeat the jointly estimated specifications using the log of PDs as the dependent variable in Columns 7 and 8. In the log specifications the public interaction is statistically indistinguishable from zero. We repeat all of this analysis using the Tier 1 gap measure in Internet Appendix Table IA5 and find largely similar results, but the public interaction is negative and statistically significant even in the log specifications.

The analysis of credit-level heterogeneity shows that the sensitivity between PD and bank capital varies across types of credits. Drawn credits, but not necessarily larger credits, are more likely to have lower PDs for capital constrained banks. In addition, riskier credits are more likely to receive lower PDs from capital constrained banks. Lastly, credits to public borrowers are less likely to exhibit sensitivity between capital constraints and PDs.

2.5 Loan spreads, risk metrics, and capital constraints

In the remainder of the paper we use loan prices to further gauge the information content of banks' risk estimates. There are two mechanisms by which bank behavior can influence PDs, each with distinct implications for estimates from loan pricing models. The first mechanism is a proportional bias in which banks

consistently report lower PDs relative to the price of the loan. The second is a selective bias that reduces the quality of PDs by making them noisier relative to loan spreads. To evaluate these two hypotheses for low and high capital banks, we compare their pricing models by examining the coefficients on PD and the explanatory power of the pricing models.

If a bank's PDs are consistently biased proportional to their level, then estimating the pricing model will result in a PD coefficient that is higher for banks with downward-biased PDs. In other words, they charge *more* per unit of risk because the spread is on average high relative to the PD they report. A core concern with internal estimates is that banks can exercise discretion over the private information incorporated in risk estimates and that this information is less complete than it otherwise could be. In this scenario, PDs will be noisier because some estimates are biased and others are not and as a result the relationship between spreads and PDs will be attenuated. Hence, these two mechanisms are at odds with regards to the PD coefficient.

To disentangle these two forces, we consider how well PDs explain loan pricing decisions. To test this, we compare the explanatory power of PD-based loan pricing models across banks. Low explanatory power relative to other banks implies that spreads are chosen using information that is not reflected in risk estimates, consistent with low-quality estimates that incorporate less relevant information.

We obtain loan pricing information from the Dealscan database. We measure the loan price using the all-in-drawn spread over LIBOR.²³ We focus the analysis on agent banks in the first quarter after the DealScan origination date. Agent banks' PDs should be particularly informative for pricing as they are the primary lead bank which typically sets the initial price of the loan. Recall, we demonstrate the relation between bias and capital is robust to a subsample of lead banks and the initial period after origination, Tables 6 and 7.

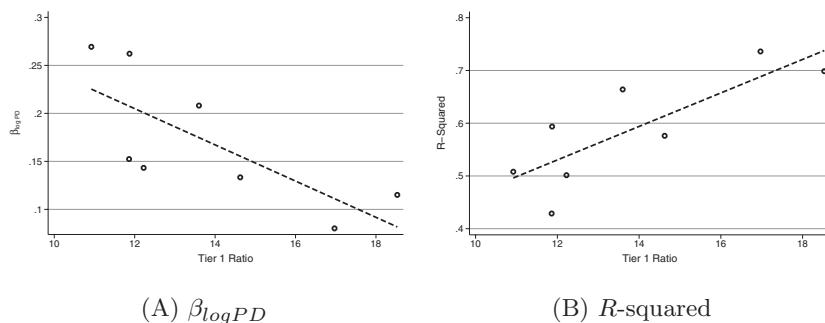
After merging the SNC data with DealScan, we obtain a sample of 4,683 loans. Seven banks have fewer than 50 loans, limiting our ability to estimate a robust pricing model. For the remaining eight banks we estimate the following regression for all first quarter, agent banks with a reported PD less than 10%.²⁴

$$\log(\text{Spread}_{j,t}) = \alpha + \beta_{\log PD} \log(PD_{j,t}) + \beta_{LC} \text{LoanControls}_{j,t} + \tau_t + \varepsilon_{j,t} \quad (5)$$

There is a distinct nonlinearity between PD and spread over this range, therefore it is important to use a nonlinear empirical model. The coefficient on the log of PD, $\beta_{\log PD}$, can be interpreted as the elasticity of spread with respect to probability of default. We control for the following loan characteristics: type (revolver vs. term loan), maturity, log commitment size, the number of

²³ For other studies that investigate loan pricing using LPC data, see Santos and Winton (2008), Hale and Santos (2009), and Santos (2011).

²⁴ The PD restriction leaves out 113 credits. We exclude these so that unusually high PDs do not affect our pricing model estimates.

**Figure 4****Bank pricing estimates versus the Tier 1 ratio**

This figure plots estimates from eight bank pricing regressions versus their average Tier 1 ratio. The pricing models regress log credit spread on credit characteristics and log PD for each bank. Figure 4A is an illustration of the elasticity of spreads with respect to PD based on estimated coefficients for each bank. Figure 4B is an illustration of the R -squared of each bank's pricing regression.

participants and other loan features including whether the loan is secured and whether the loan contains dividend payment covenants. We also include time fixed effects to account for macroeconomic factors. Standard errors are robust to heteroscedasticity and clustered by borrower. The results of these regressions can be found in Internet Appendix Table IA6.

All else equal, the coefficient on PD should be higher for banks with downward biased PDs. Given the relation between bias and capital documented earlier, we expect higher betas for those banks with lower capital. Indeed, a plot of betas shows a downward relation where capital is measured by Tier 1 gap. The more constrained a bank, the greater the elasticity with respect to PD (Figure 4A).

In addition, we expect banks with less informative PDs to have a lower R -squared. If bankers do not consistently include information in PDs, the PDs will have less explanatory power with regard to prices. Figure 4B demonstrates a distinct positive slope between the R -squared of the pricing equation and Tier 1 gap; that is, lower capital banks have less informative PDs. Despite being limited in our ability to make statistical inferences because of the limited number of banks, both figures imply distinct linear relations consistent with our hypothesis that low capital banks report lower PDs that contain less information about the borrower.

One drawback of estimating pricing coefficients at the bank level is that the pricing coefficients are premised on different sample sizes. As a robustness check, we repeat the estimation by forming roughly equally sized portfolios of credit-quarters based on deciles of the Tier 1 ratio. Using these deciles, we estimate ten sets of coefficients and R -squareds from Equation (5). See Internet Appendix Table IA7 for the pricing estimates and Figure IA2 for the illustrative relation with the Tier 1 ratio. The relation between implied spread and capital is negative but not as striking like in Figure 4A. The relation between R -squared

and capital decile is positive. Both results are consistent with attenuation in the PD coefficient as a result of less informative PD estimates.

The positive correlation between *R*-squared and Tier 1 measures suggests that low-capital banks' loan prices are determined by information that is not incorporated in PDs. Higher Tier 1 capital banks, on average, report PDs that are more informative for loan prices relative to low Tier 1 banks, consistent with more information being incorporated in their reported risk metric. These results are consistent with a regulatory motive and are difficult to reconcile with the alternative hypotheses related to risk perception, risk tolerance, credit selection, or information advantages.

3. Conclusion

Using a novel data set of syndicated loan participants and the internally generated risk metrics they report for regulatory purposes, we identify systematic cross-sectional variation in how banks rate common credits. We find that the variation in probability of default is strongly correlated with measures of a bank's Tier 1 capital ratio. On average, banks with lower capital report lower estimates relative to the average reporting bank in the syndicate. The magnitude of these differences is meaningful when we aggregate credits into portfolios, resulting in differences in risk-weighted assets as large as 20%.

We explore several explanations for this correlation and cannot dismiss the concern that banks under-report risk estimates relative to their peers in response to capital constraints. Low capital banks are more likely to report lower risk for risky and drawn credits, findings which are consistent with a regulatory motive. More importantly, low capital banks appear to concentrate this biasing behavior on private firms where they are more likely to have greater discretion over the inputs to their risk models. Further analysis suggests that low capital banks not only have lower risk estimates but that they are less informative. Low capital banks set spreads on loans that are less consistent with their reported PD, a fact that is difficult to reconcile with alternative explanations.

Our findings highlight the potential dangers of self-reported risk metrics. Banks that are already relatively fragile report commonly held credits are less risky than their peers, a finding that implies that low-capital banks are even more risky than they appear. Our analysis suggests this is consistent with a regulatory arbitrage motive, but regardless of the underlying motive, systematically disparate risk estimates present a distinct challenge to an effective regulatory regime premised on self-reporting.

Our results present supporting evidence for the adverse effects suggested by the Lucas critique. Given the increasing reliance on bank-generated information, we highlight the critical role of incentives on the quality of these disclosures. This conclusion echoes concerns espoused by Federal Reserve Board Governor Daniel Tarullo, who noted in 2014 that "the combined complexity and opacity of risk weights generated by each banking organization

... create manifold risks of gaming, mistake, and monitoring difficulty.” To ensure the integrity of internally generated risk estimates, new programs should include mechanisms that monitor information quality and incentivize the production of accurate risk metrics. Lastly, our findings also lend credence to the use of a leverage ratio that caps gains from internally generated risk estimates.

In this paper we have exclusively focused on ex ante comparisons between banks and their peers. We do not compare the accuracy of risk estimates by considering ex post loan outcomes as risk estimates are what determine regulatory capital. However, the overall accuracy of risk estimates is an interesting area for future research. Our preliminary investigation of this issue, based on loan performance and secondary loan market pricing, did not yield robust results. It is possible this derives from the lack of economic volatility in our sample period.

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